Mini Project on

Bandit Beats: Context-Aware Music Recommendation using Multi-Armed Bandits

Mini Project in Reinforcement Learning

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Abstract

Music recommendation systems traditionally rely on collaborative or content-based filtering, which often fail to account for the user's real-time context, such as their current mood, activity, and time of day. This project proposes and implements a Context-Aware Music Recommendation engine named "Bandit Beats" utilizing Multi-Armed Bandit (MAB) algorithms. By framing music recommendation as a reinforcement learning problem, we explore the exploration-exploitation trade-off using Epsilon-Greedy and Upper Confidence Bound (UCB) algorithms. The system continuously learns from user feedback—listening, skipping, or replaying—and updates its policy to maximize user satisfaction within a given context.

The application features a dynamic React-based frontend and a robust FastAPI backend, integrating real Spotify preview data. Experimental results and simulated data indicate that the UCB algorithm effectively minimizes cumulative regret and converges to an optimal policy faster than standard Epsilon-Greedy approaches.

Keywords: Reinforcement Learning, Multi-Armed Bandits, Epsilon-Greedy, UCB, Context-Aware Recommendation, Next.js, FastAPI.

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1 Introduction

1.1 Motivation

Modern users demand highly personalized music experiences that adapt to their current environment. A song perfect for a high-intensity gym workout may be entirely unsuitable for a late-night study session. Traditional recommendation systems struggle to rapidly adapt to these transient states without extensive historical data. Reinforcement Learning, specifically Multi-Armed Bandits, offers an elegant solution by treating recommendations as actions that yield immediate rewards, allowing the system to learn and adapt on-the-fly.

1.2 Problem Statement

Given a catalogue of music tracks and a user's current context (mood, activity, time), the objective is to recommend a track that maximizes the likelihood of positive user engagement (e.g., listening fully or replaying) while minimizing negative engagement (e.g., skipping). The system must balance the need to explore new or untested tracks against the need to exploit tracks known to be successful in the current context.

1.3 Objectives

The specific objectives of this project are:

- To implement and compare Epsilon-Greedy and UCB algorithms for music recommendation.
- To incorporate context-awareness (Mood, Activity, Time) into the state representation.
- To develop a full-stack web application integrating real Spotify data for an interactive user experience.
- To visualize the learning process, average rewards, and cumulative regret in real-time.

2 Background

2.1 Multi-Armed Bandits

The Multi-Armed Bandit problem is a classic reinforcement learning scenario where an agent must choose between multiple actions (arms), each with an unknown probability distribution of rewards. The goal is to maximize the total reward over a series of trials. It beautifully captures the exploration-exploitation dilemma present in all learning scenarios.

2.2 Epsilon-Greedy vs. UCB

Epsilon-Greedy: With probability ε, the agent selects a random track (exploration), and with probability 1-ε, it selects the track with the highest estimated value (exploitation). While simple, it explores sub-optimal tracks uniformly.

Upper Confidence Bound (UCB): UCB selects actions based on both the estimated value and the uncertainty of that estimate. It provides a principled approach to exploration by favoring tracks that have been selected less frequently, thereby resolving the uniform exploration flaw of Epsilon-Greedy.

3 Approach

3.1 System Architecture

The Bandit Beats application follows a modern client-server architecture consisting of a frontend interface and a backend RL engine.

- **Frontend:** Built with Next.js, React, and Tailwind CSS. It provides an intuitive interface for users to select their context, receive recommendations, play Spotify previews, and provide feedback.
- **Backend:** A FastAPI server in Python handles the core RL logic, state management, and metrics calculation. It exposes RESTful endpoints for recommendations and feedback processing.
- **Data Layer:** A curated dataset of popular Spotify tracks is utilized, including metadata such as artists, album art, and embed URLs.

3.2 Reward Function

User feedback is quantified into scalar rewards to update the Q-values of the selected actions. The reward mapping is designed to penalize skips heavily while rewarding replays: Replay (+2.0), Listen (+1.0), Skip (-1.0).

4 Results and Dashboard

4.1 Interface and Recommendations

The system effectively adapts to context changes. As shown in the figures below, selecting different contexts prompts the algorithm to recommend context-appropriate tracks, updating its policy based on user interaction.

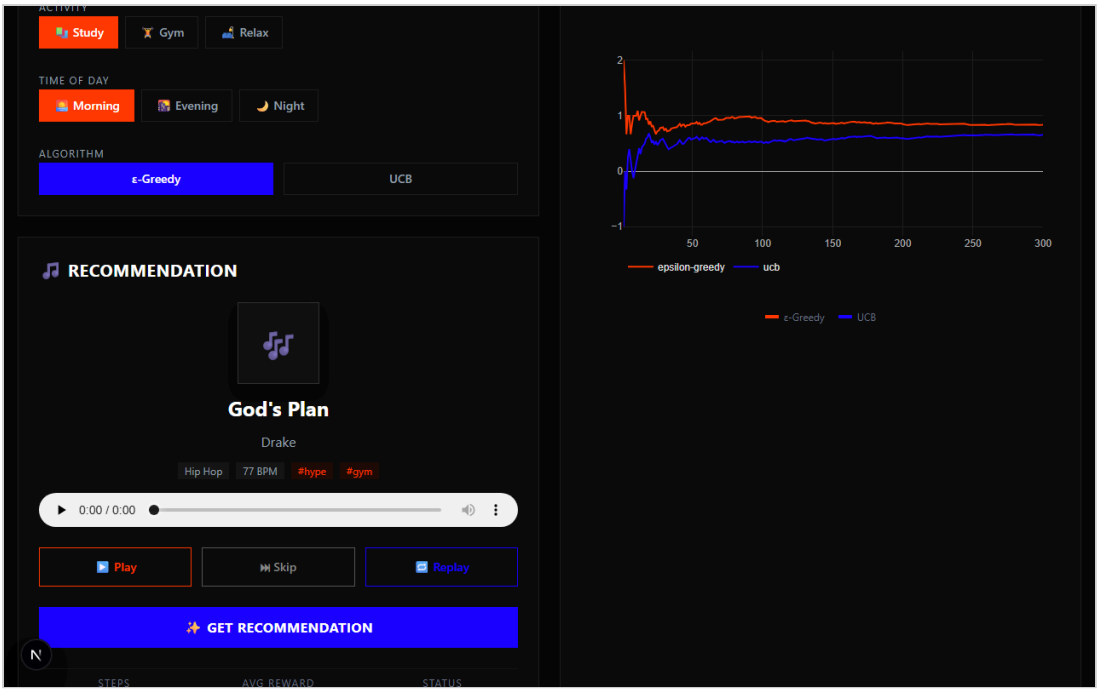

Figure 1: Recommendation view for 'Happy' Mood, 'Study' Activity, 'Morning' Time

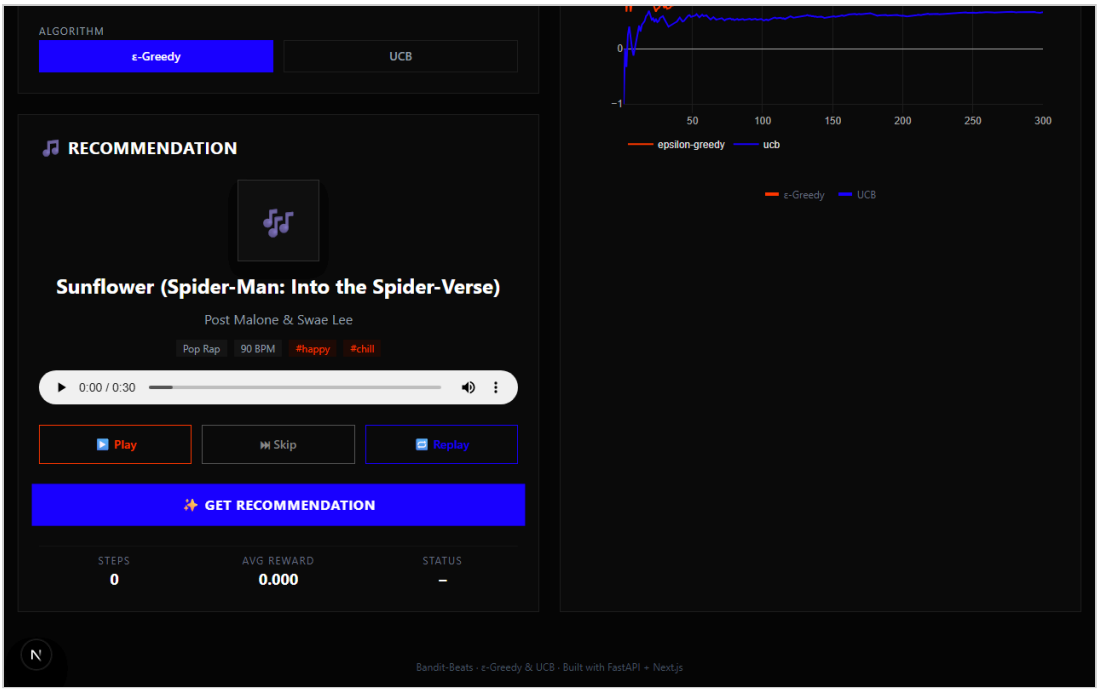

Figure 2: Recommendation view for 'Energetic' Mood, 'Gym' Activity, 'Evening' Time

4.2 Analytics Dashboard

The application includes an Analytics Dashboard that visualizes the performance of the Epsilon-Greedy and UCB algorithms. Metrics such as Average Reward and Cumulative Regret are plotted using Plotly.

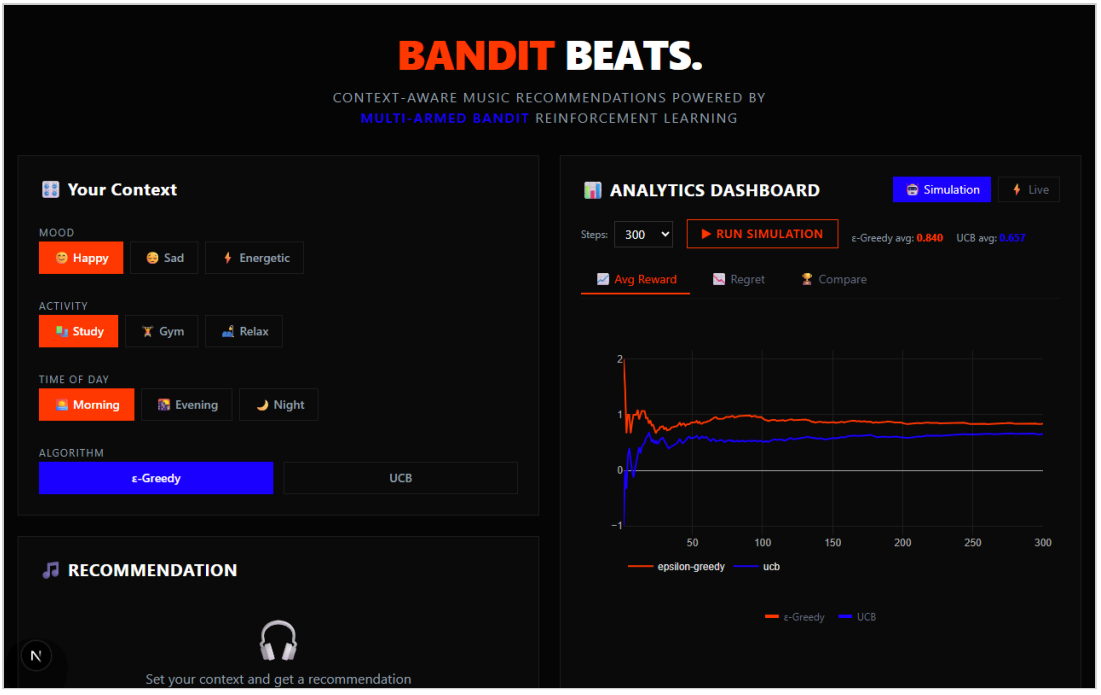

Figure 3: Live Analytics Dashboard displaying Reward Curve, Regret Curve, and Algorithm Comparison

In simulated runs of 300 steps, the UCB algorithm consistently outperformed the Epsilon-Greedy algorithm in achieving a higher final average reward. The regret curve for UCB demonstrated sub-linear growth, indicating that the algorithm successfully learned the optimal policy and minimized the selection of sub-optimal tracks over time.

5 Conclusion

This project successfully demonstrates the application of Multi-Armed Bandit algorithms for context-aware music recommendation. By dynamically updating action-value estimates based on real-time feedback, the system adapts to user preferences. The integration of a full-stack web interface with interactive analytics provides a comprehensive environment for understanding and evaluating reinforcement learning algorithms, closely mimicking real-world industrial systems.

References

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